

be in increasing order).

2.1.4. Measured Value

This section provides requirements on the attributes MeasuredValue, MinMeasuredValue, MaxMeasuredValue. Accuracy of MeasuredValue is discussed in the following section.

2.1.4.1. Constraint

Where MinMeasuredValue or MaxMeasuredValue attributes are mandatory the null value MAY be used to indicate that a limit is unknown.

For any measurement cluster with MeasuredValue, MinMeasuredValue and MaxMeasuredValue attributes, the following SHALL be always be true:

- If MinMeasuredValue and MaxMeasuredValue are both known, then MaxMeasuredValue SHALL be greater than MinMeasuredValue.
- If MaxMeasuredValue is known, then MeasuredValue SHALL be less than or equal to MaxMeasuredValue.
- If MinMeasuredValue is known, then MeasuredValue SHALL be greater than or equal to MinMeasuredValue.

2.1.5. Measurement Accuracy

Measurement clusters MAY express the accuracy of their measurements with a [Tolerance](#) attribute expressing a simple magnitude of error, or with a MeasurementAccuracyStruct expressing magnitude or percentage ranges of error for different ranges of measured values.

2.1.5.1. Tolerance Attribute

For any measurement cluster with a MeasuredValue and Tolerance attribute, when Tolerance is implemented the following SHALL always be true:

- The Tolerance attribute SHALL indicate the magnitude of the possible error that is associated with MeasuredValue attribute, using the same units and resolution. The true value SHALL be in the range (MeasuredValue – Tolerance) to (MeasuredValue + Tolerance).
- If known, the true value SHALL never be outside the possible physical range. Some examples:
 - a temperature SHALL NOT be below absolute zero
 - a concentration SHALL NOT be negative

2.2. Illuminance Measurement Cluster

The Illuminance Measurement cluster provides an interface to illuminance measurement functionality, including configuration and provision of notifications of illuminance measurements.